

FALCON's Response to Style Union NEO (3D ASRS System)

September 25, 2025

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

Mr. Sankatmochan

Offer Ref: F24-00438C_Ver 1; Date 25.09.2025

Subject – Techno commercial Offer for Style Union for NEO

Dear Sir,

We are pleased to submit our Techno-Commercial Offer to **Style Union** in response to your requirement for **NEO ASRS system**.

We are proud to emphasize that Falcon Autotech, serving as the prime contractor for this project, is a global leader in intra-logistics automation solutions. The company boasts an impressive portfolio, with over 200 Installations deployed worldwide, with handling capacities from 800 to 60,000 Units/Hour. Falcon Autotech has an outstanding track record in the design, manufacture, and installation of intralogistics automation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops, visits, and calls with your different teams, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

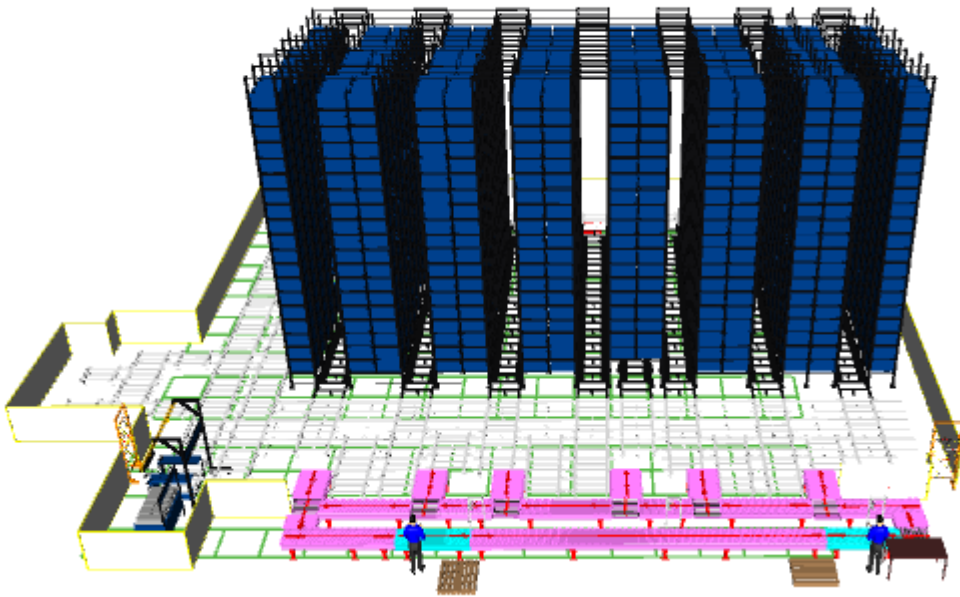
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the requirement, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response To Style Union's requirement for NEO ASRS System



Falcon Autotech Private Limited

**Plot No. 87, Sector Ecotech-1, Extension-1, Greater Noida, Uttar
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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PS	Pouch Sorter
3	GTP	Goods to person
4	PPH	Parcels Per Hour
5	AC	Alternating Current
6	DC	Direct Current
7	PLC	Programmable Logic Controller
8	IT	Information Technology
9	BOQ	Bill Of Quantity
10	I/O	Input/ Output
11	PDP	Power Distribution Panel
12	PC	Personal Computer
13	UPS	Uninterrupted Power Supply
14	MTS	Multi-Tier Shelving
15	PTL	Pick/Put to Light

2. Executive Summary

Falcon is pleased to confirm its interest in responding to the automation requirement presented by Style Union. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening to and understanding your needs and ensuring this project's success. As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the said system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following inputs provided by the customer:

- **The NEO – ASRS system running with 16 NEO Bots & 2694 Bin Locations.**
- **Total 2 Goods to Conveyor Stations – 1 Nos for picking, 1 Nos for put away.**

A tailor-made and simple layout, specifically designed for style Union. The proposed layout is the result of the technical requirements listed during our various meetings and E-mail exchanges.

1. Falcon's reliable Parcel sortation system

These systems are globally being used by most innovative brands such as Amazon, Asendia, Fastway, Flipkart and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

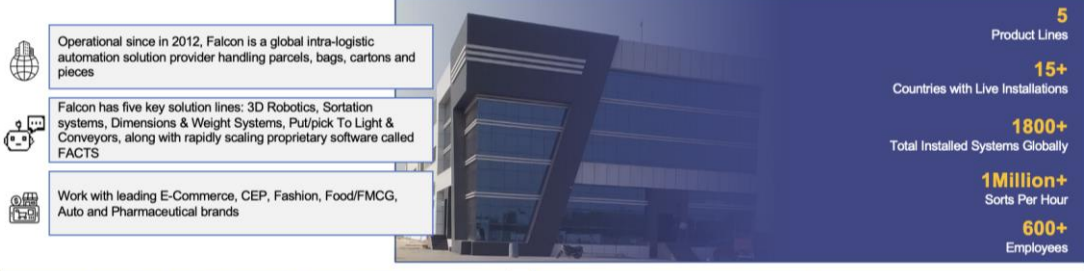
2. Commitment to Quality Systems

Demonstrating Falcon's clear commitment to the Style Union Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from the completion of Installation & commissioning.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1 Million+ Sorts Per Hour

600+ Employees

Long-standing Relationships with industry leaders



















News & Articles


FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
 Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

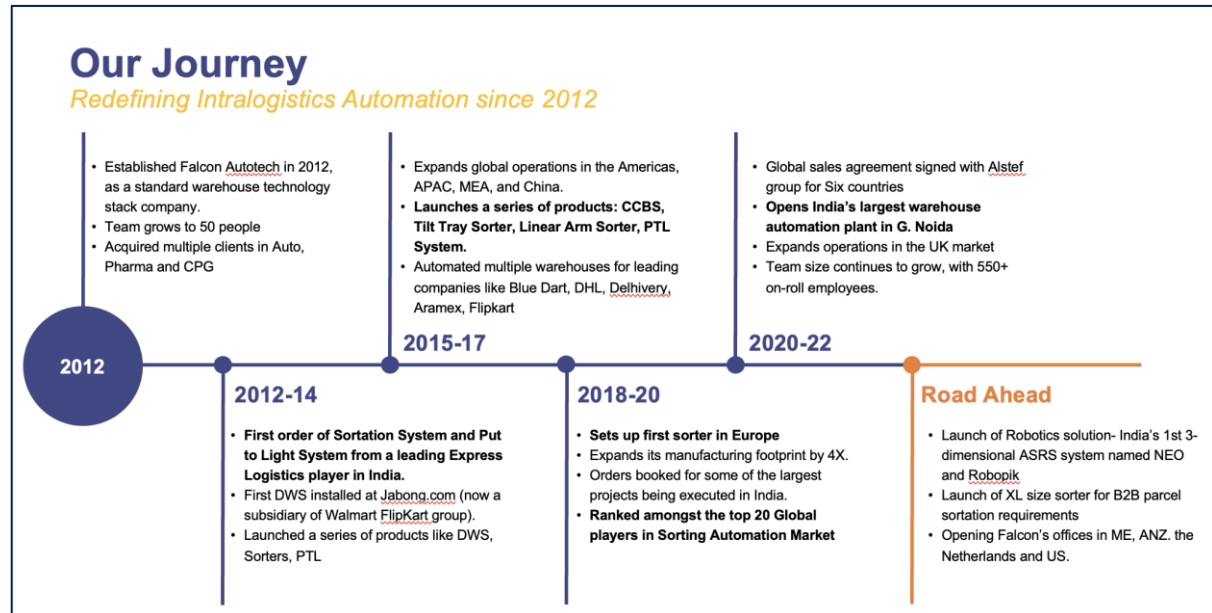

FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
 Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
 Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation companies; our vision is to become the top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world-class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Pop-up Sorter
- Sweep Sorter
- Pusher Sorter



PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
- R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
- Mini (600MM)
- Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

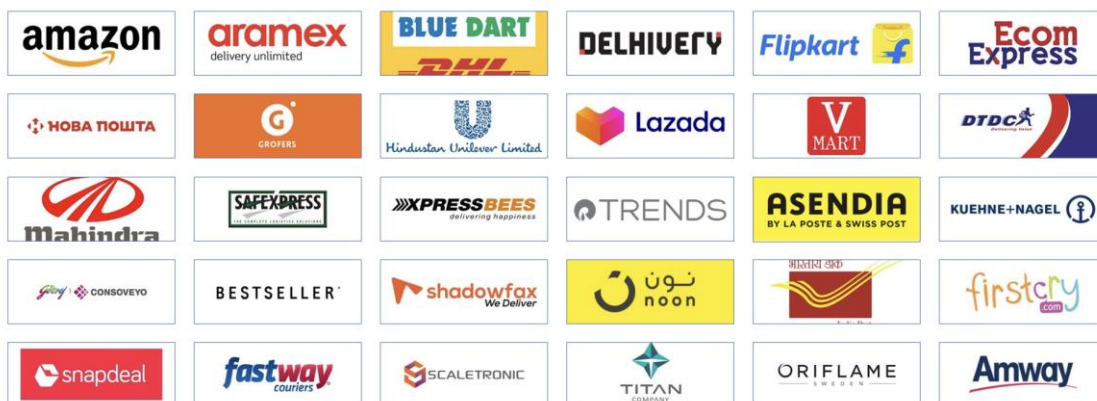


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of the above product lines for effective materials handling, sortation, and movement. The process is controlled in real-time by our in-house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions, and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long-standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

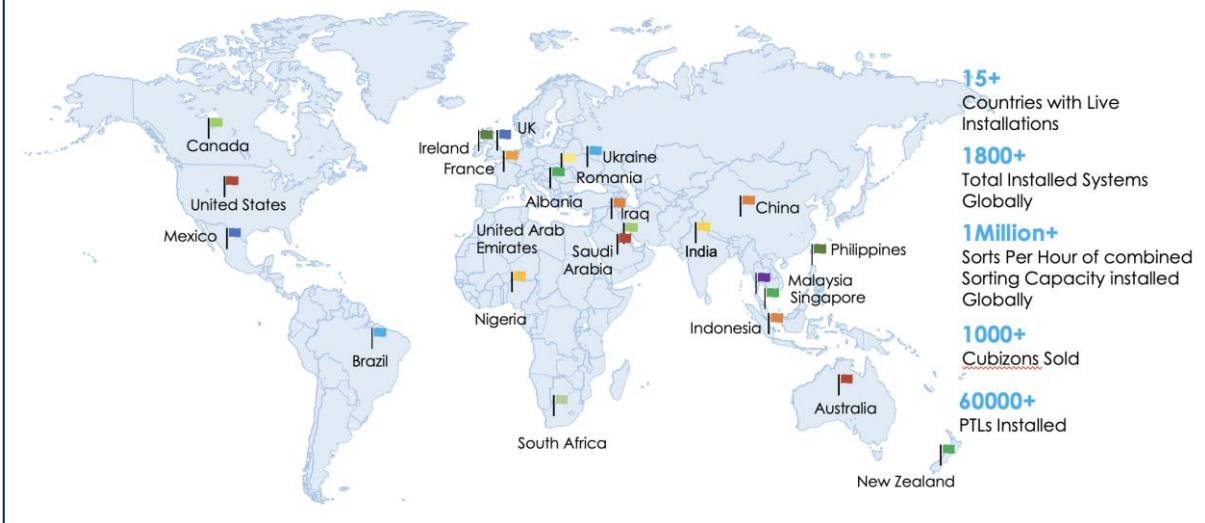
Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements



With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

Global Market Presence



Falcon's Experience and Achievements in Warehouse Automation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide. Only company to be able to offer a **Fully Integrated AMS**



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FALCON AUTOTECH

Falcon Autotech expands its wings by opening its office in The Netherlands, Europe

April 15, 2024 03:32 ET | Source: [Falcon Autotech Private Ltd](#)

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NEW DELHI, India and AMSTERDAM, April 15, 2024 (GLOBE NEWSWIRE) -- Falcon

Autotech (Falcon), a leading global intralogistics automation solutions provider, opens its



Breaking Barriers and Optimizing Efficiency – Journey of Delivering India's Largest Sortation System

"It was a bold decision. If it didn't work, our company would have shut down," says Naman Jain, founder and CEO of warehouse automation startup Falcon Autotech, referring to a big gamble he had taken more than a decade ago. The year was 2013, Jain and his team were negotiating a deal with a major e-commerce logistics company that had just placed an updated order for a sorter with a capacity of 6,000 parcels per hour. That was 4x the specification earlier agreed upon.

"All the founders came together, and we took a calculated risk," recalls Jain. At the time, Falcon did not have the technology to develop such sorters, and hence, the said client refused to pay any money upfront and offered to pay the machine's price in monthly installments. However, there was a caveat: if it failed even once, Falcon would simply take the machine back and refund the entire amount. Falcon's 'calculated risk' paid off. The machine worked flawlessly and was handed over to the parcel company after 36 months. "We are now doing projects worth INR100 crore. That was unthinkable 10 years back," says Jain.

Falcon Autotech has covered a lot of ground since that big bet. Last month, it installed India's largest sortation equipment at



Economic Times Features Falcon Autotech

Naman Jain
Chief Executive Officer

From sorters to conveyors & robot based systems, here's the top tech warehouses are investing in

India's warehousing industry has travelled a long distance from "godowns" to modern storage facilities called Grade A warehouses.

Warehouse automation is a gradual process. India started off late, but that may actually help it bypass some of the mid-age technologies and adopt the latest ones. While several players are still warming up to automation, which are the



Falcon Announces Successful Go-Live of Automated Parcel Sortation Solution at DTDC's Chennai Facility

New Delhi, India – 2nd Aug 2023: Falcon Autotech, a leading supplier of intralogistics automation solutions, has been selected by DTDC Express Ltd, one of India's leading integrated express logistics company, to automate its parcel sorting operations at its super hub of 1,75,000 sq ft in Chennai, Tamil Nadu. Using its cross belt sorter technology, Falcon has designed DTDC's parcel sorting system, which can handle 6,000 parcels per hour, operate in a 24 X 7 environment, and can be expanded to cater to future growth.

The new linear cross-belt solution leverages cutting-edge technology to automate key sorting processes in DTDC's warehouse, including parcel profiling and sorting. This solution is designed to optimize the space requirements for sorting operations, increase efficiency, and reduce operational costs.

"We are thrilled to see our warehouse automation solution go live with DTDC, this solution is a testament to our commitment to providing innovative intra-logistics solutions that meet the evolving needs of our customers," said Falcon's CEO Naman Jain.



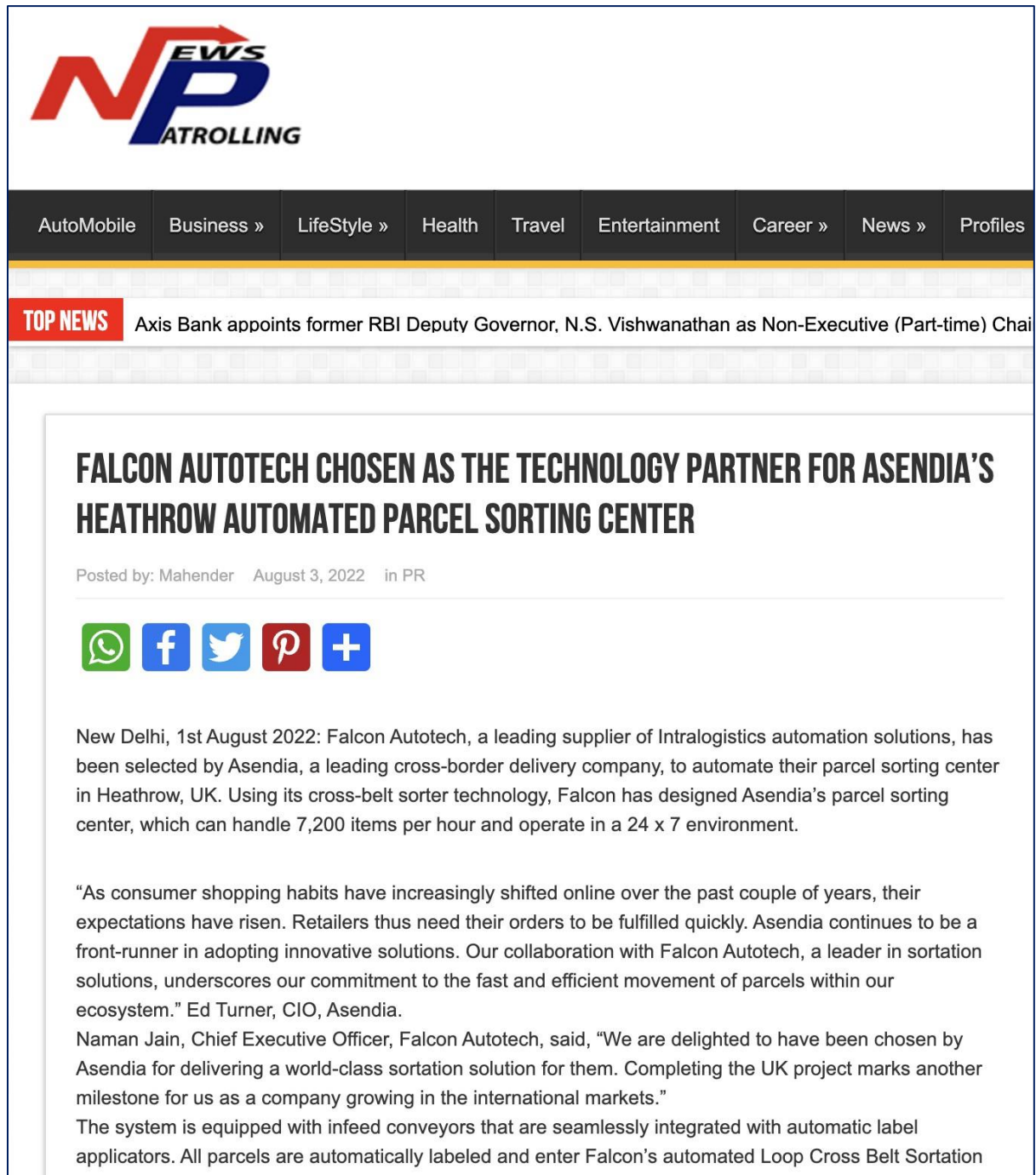
Transforming Indian Retail: The Impact of ASRS on Warehousing Operations

In recent years, the Indian retail industry has experienced unprecedented growth, propelling the nation to become the fifth-largest global retail destination. The industry, marked by its resilience and adaptability, went through significant transformations during the pandemic. While online shopping saw a remarkable surge, the reopening of physical stores ushered in a resurgence of the multi-sensory shopping experience. As a testament to this growth, shopping malls now encompass an astonishing 23.25 million square feet of retail space. However, the evolving consumer preferences have placed substantial pressure on businesses, necessitating the seamless integration of e-commerce and in-store experiences, which, in turn, has led to complex logistics challenges. The traditional warehousing systems struggled to cope with the demand for faster processing and the requirement for expanded storage capacities.

Historically, retailers had heavily relied on manual labor for their warehousing operations. However, the dynamism of today's retail market demands a shift towards automation. Enter Automated Storage and Retrieval Systems (ASRS), a technological

4. Falcon's Experience and Achievements in Sortation Space Globally

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- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Riyadh)

In 2019, customers selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

The solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. The system is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

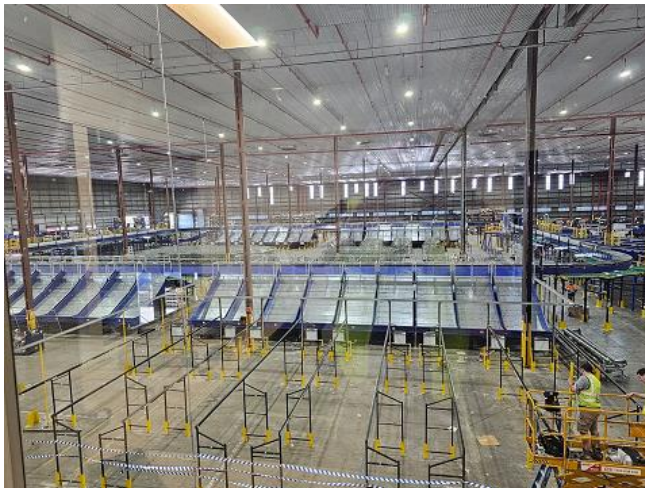
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while Sub-System 2 is designed to handle Small E-commerce packages.

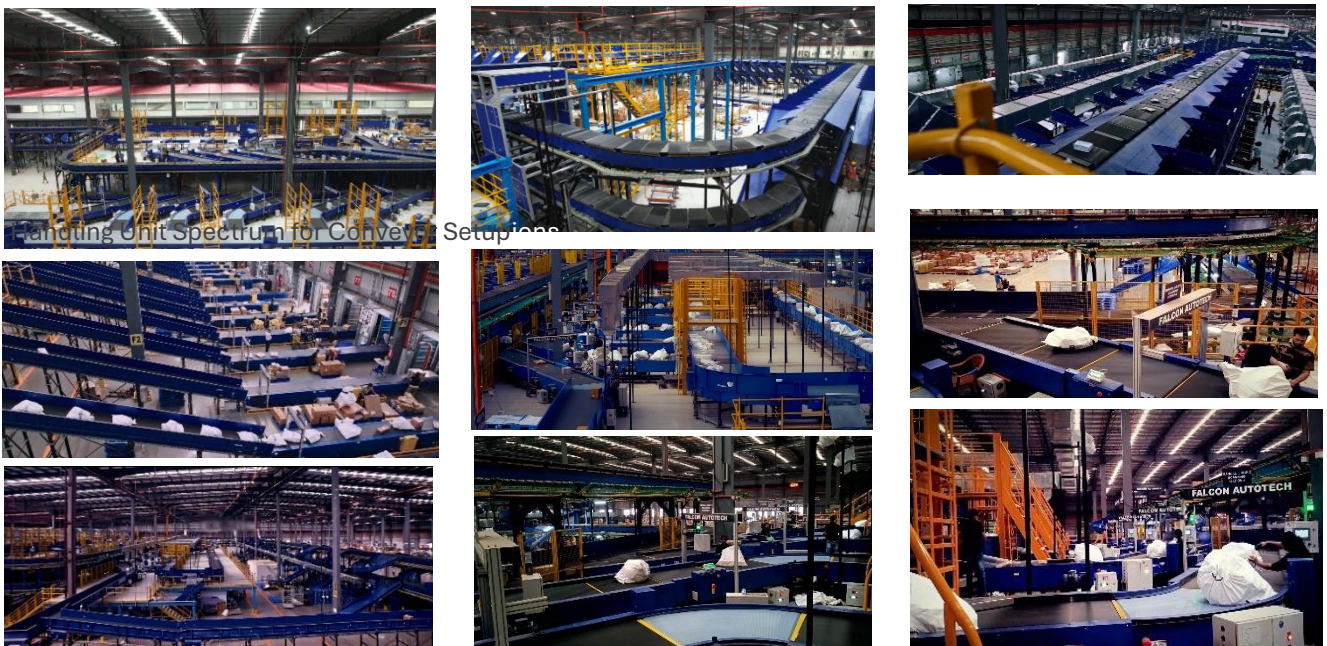
Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- 2 Sets of Double Decker CBS Sorters
- Mezzanine Structures
- Automated Singulators
- Fully & Semi-Automatic Inductions with weighing conveyors
- Telescopic belt conveyors
- PVC Belt Conveyors
- Modular Belt Conveyors
- Spiral Chutes with Braking rollers
- 5-Sided Scanning Tunnels
- Direct Bagging Chutes
- Put to Light Chutes
- Volume Distribution systems
- High Availability Server Systems
- WCS

Site Pictures –



5.6 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

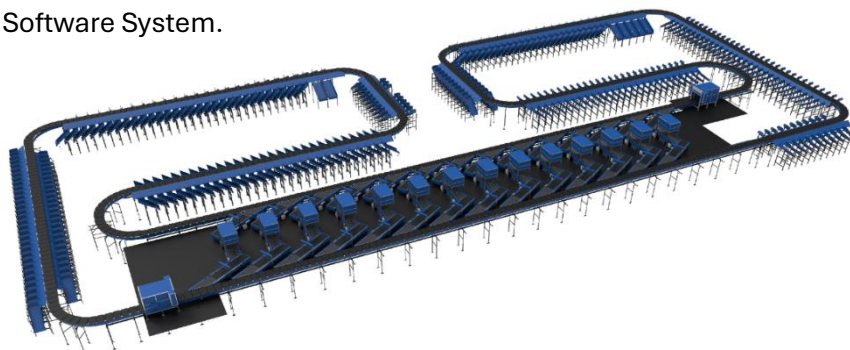
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq. Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Pre-sort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.



5.7 Project 6- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

The customer chose Falcon Autotech based on its unique design, capability of seamless integration with WMS and life cycle support services.

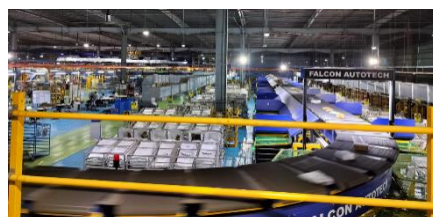
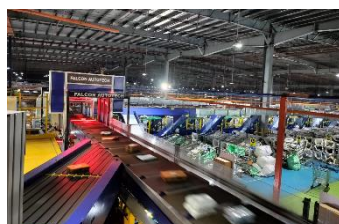
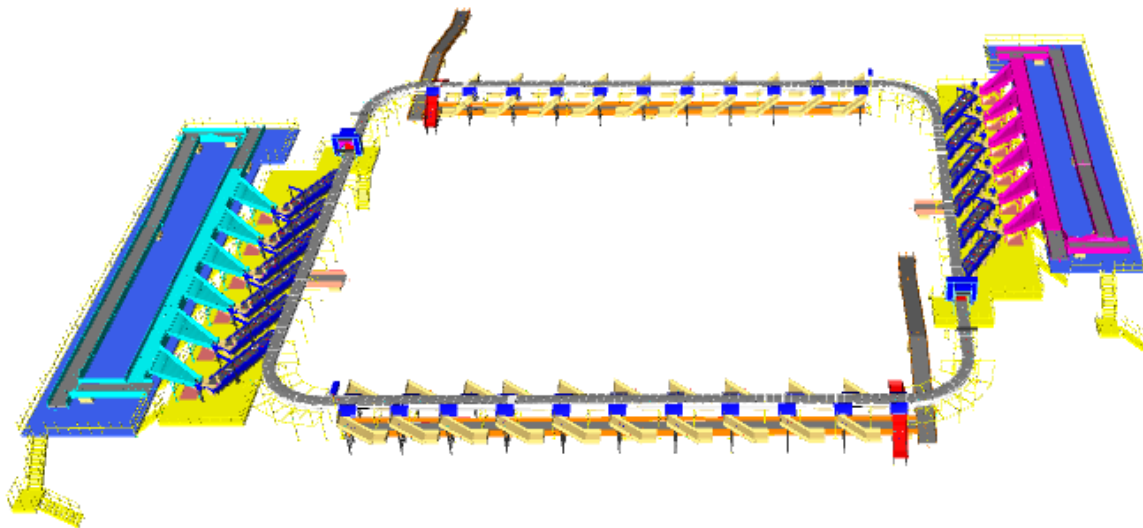
Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 40 Collection Type Chutes

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Collection type chutes
- Bag take out system.
- WCS Software System.

Layout and Site Pictures -



6. Handled Product Spectrum

6.1 Product Spectrum

Specification	Unit	Value
Max Length	mm	600
Max Width	mm	400
Max Height	mm	300
Max Weight	Kg	40
Product Type	type	Bins, Boxes, Trays, etc

6.2 Shipments to be loaded shall have the following characteristics:

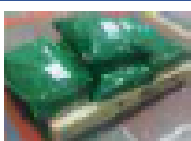
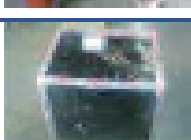
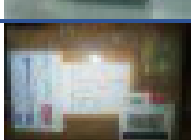
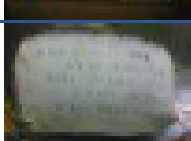
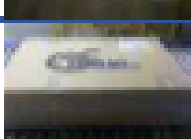
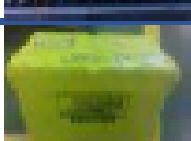
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behaviour of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or
damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and
referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Shipments not loadable

All products that are not within the range described here are considered non-conveyable products and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Conveyable Shipments

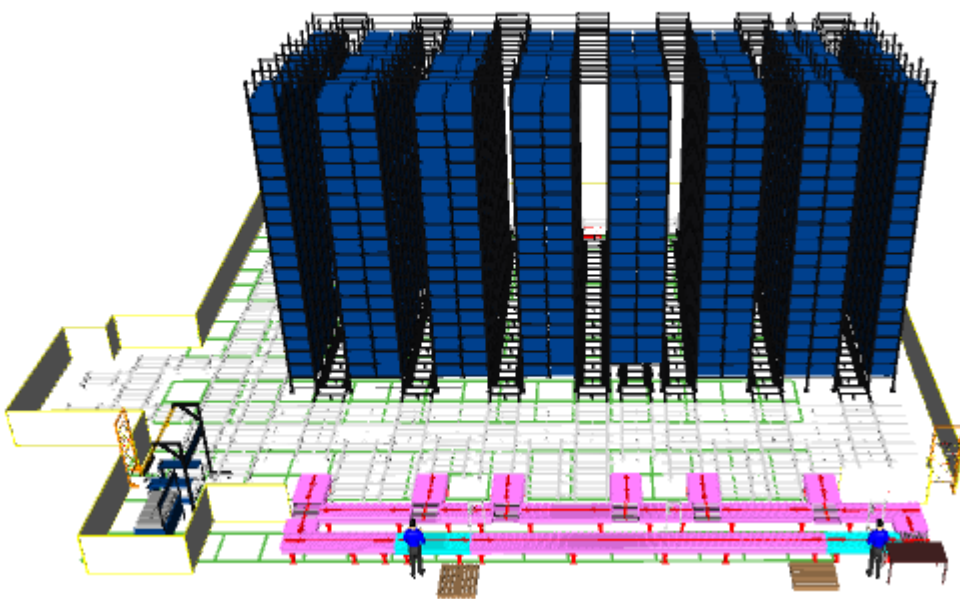
PARCEL TYPE	DESCRIPTION
	Strongly Strapped A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item A cardboard parcel coated in a packing paper.
	Cloth Covered Item Often import parcels that are of irregular shape.
	Wine Box A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box A cardboard parcel is often the norm.
	Totes A large plastic, lidded box.
	Flats A parcel that often contains documents.
	Poly Bags It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, integration with Style Union group's WMS and acceptance testing of the complete automation system comprising of Conveyor Automation to handle various up-stream and down-stream operations.

7.2 Summary of the system



Process flow of the system:

1. Inbound (Put-away Process)

- a) Boxes/Totes will be received at the Goods-to-Conveyor (GTC) stations after preliminary processing (P&A). Operators are stationed at the GTC to manage induction.
- b) The operator initiates a Put-away Wave in the system. Upon initiation, a Neo bot arrives at the GTC station to collect the Box/Tote.
- c) The operator scans the Box/Tote barcode using the scanner integrated with the GTC station. The Box/Tote is then placed on the GTC station, from where it is transferred by Neo into the ASRS storage location.
- d) Once put-away is completed successfully, the Neo bot returns to the GTC station to repeat the process until all assigned put-away tasks are completed.

2. Outbound (Picking & Dispatch Process)

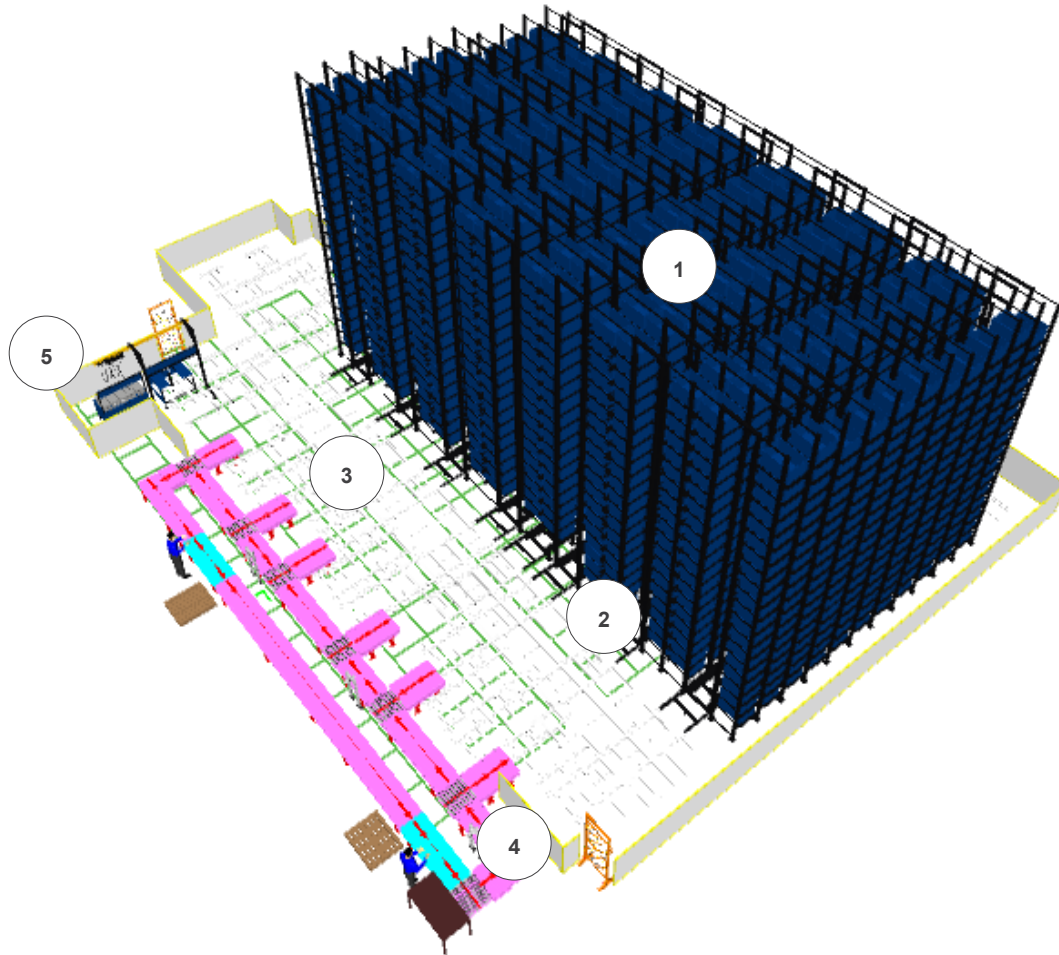
- a) The operator initiates a Pick Wave based on the dispatch plan or dispatch list received from WMS.
- b) Once the pick wave is triggered, Boxes/Totes are retrieved from ASRS storage and delivered to the GTC station sequentially, as per the dispatch list.
- c) Operators pick the required items as directed by the system and prepare them for further downstream activities (sorting, packing, or direct dispatch).

7.3 Main Benefits of the NEO ASRS System

Falcon's Automated Storage and Retrieval System (ASRS) NEO offers a multitude of benefits over traditional storage systems, revolutionizing the way goods are stored and retrieved within warehouses.

- **Enhanced Productivity:** NEO enhances your operator productivity by 4-5X. By eliminating the wait time, walk time and search time from picking operations.
- **Space Optimization:** NEO storage grid can go up to 16 meters to maximize vertical space utilization in your warehouse, allowing you to store more products in a smaller footprint and significantly increasing your warehouse's storage capacity.
- **Fast and Accurate Retrieval:** With automated picking and retrieval, NEO ensures lightning- fast and error-free order processing. This means you can meet tight delivery schedules with confidence and exceed your clients' expectations.
- **Inventory Management:** Gain real-time visibility into your inventory levels, helping you make informed decisions.
- **Scalability:** Whether your warehouse is small, large, or somewhere in between, NEO is scalable to adapt to your evolving business needs.

8. Layout Overview



Legend

1. NEO Racks
2. NEO Bots
3. NEO Tracks
4. Goods to Conveyor Station
5. Battery Swapping Station

9. System Components

9.1 NEO Bot

NEOBot is your ultimate warehouse companion that never sleeps or is on leave. A powerful machine with minimum moving parts, NEObot is always ready to deliver at its best. Exquisitely designed and engineered, the bot requires no monitoring while at work. Using advanced Lidars and AI algorithms, it moves independently in the storage grid without continuous control from the software.

Technical Specification

Weight: 200Kgs
Payload Capacity: 40Kgs
Speed: 2m/s
Size: 1016 x 908 x 469 mm
Operating temperature: 0-40 degrees C
Juice: Swappable LiFepo4 battery



9.2 NEO Bin

Products are stored and moved in specialized bins or totes designed to be handled by NEO system. Each bin can be partitioned into 2/3/4/6/8 sub-sections for better product segregation.

Technical Specification

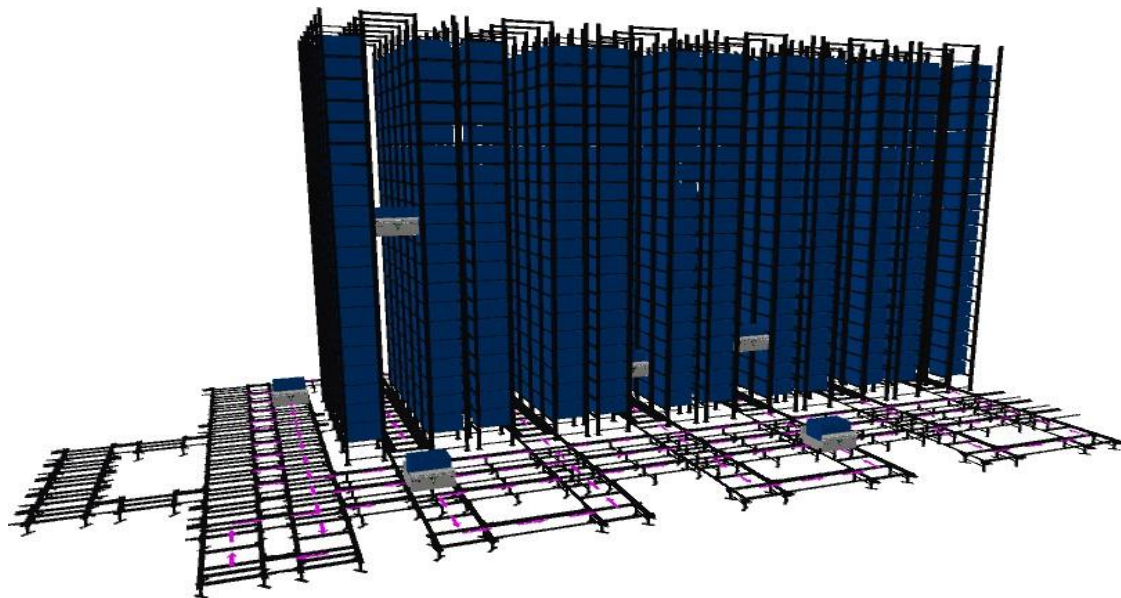
Partitions: 2/3/4/6 sub-sections
Bin volume: 5.75 cubic feet
External Dimension: 810*568*425 mm



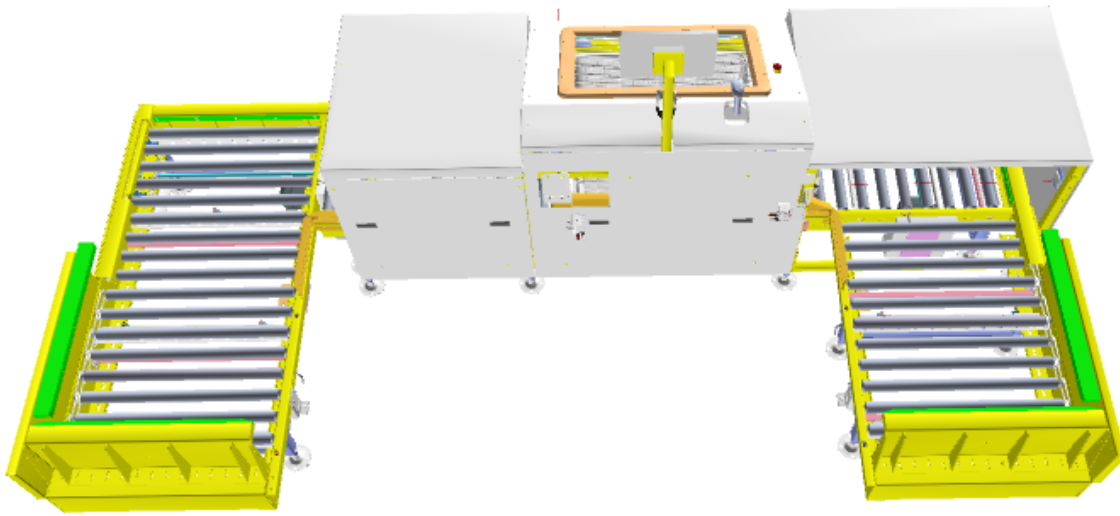
9.3 NEO Grid

NEO grid is the steel structure which stores all your goods. With carefully crafted racks and tracks, the grid provides easy access to bots for storing and retrieving bins, cartons or trays. Based on the urban city layouts, NEO grid combines interior tracks, highway tracks and superhighway tracks inside aisles for bot's movement between blocks to the grid and workstations.

When a NEObot picks up a NEO bin (which is waiting at the pick station), the system automatically assigns an appropriate rack inside the system, and NEOBots proceed to that rack position to store the chosen bin.



9.4 Goods to Conveyor Station



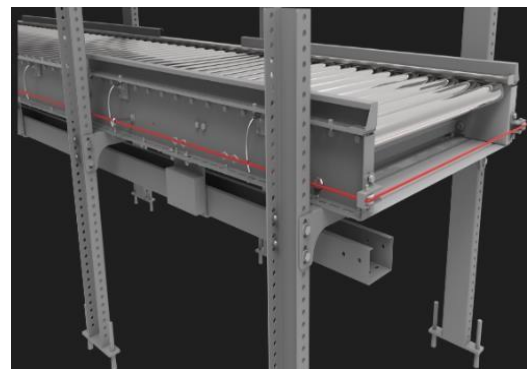
- The bins are brought to the picker by bots
- The bot drops the bin in the conveyor and proceeds for the next task
- The completed bins are carried out by the same or different bot, based on the availability
- Useful when the number of bin presentations required in a station are high
- Suitable for medium-to-large warehouses with more pick station

9.5 Accumulation Roller Conveyor

Falcon's Accumulation conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's accumulation conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	Mm	900
Conveyor Height	Mm	As per Layout requirement
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Round/Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity per unit length of Conveyor	kg/m	40
Roller Span	Mm	Approx. 600
Roller Diameter	Mm	Approx. 50
Pitch of Rollers	Mm	75
Zone Length	Mm	825
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Steel guides at 50mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

9.6 Battery Charging Station

It is the area where the battery of the shuttles is swapped and charged. NEOBots with low volume hours can be replaced with new charged batteries or straight parked for charging.



Note- Battery operational time is 6-8 hours (Fully charged), charge time is 2-2.5 hours (during non-operation hours)

10. Proposed System Technical Details/BOM

10.1 Mechanical equipment

S.NO	Component	Qty	UOM	Set
1	NEO Bots	12	Nos	1
2	Spare battery	4	Nos	1
3	Bin Position Cost (500mm Pitch)	2694	Nos	1
5	Safety Fencing	90	Meter	1
6	Charging Station	1	Set	1
7	Rescue Kit (Clear Height- Up to 12mtr)	10.5	Mtr	1
8	IT Server and IT Infra	1	Set	1
9	Mechanical and Electrical Infra	1	Set	1
10	WCS Cost	1	Set	1
11	WCS Annual License	1	Set	1
12	IT Integration	1	Set	1
13	Accumulation Roller conveyor	38.4	Meter	18
14	Pop up Unit*	7	Set	1
15	Side barcode Scanner	3	Set	1
16	PLC based Control	1	Set	1
17	Safety Doors	2	Set	1

Note: 1 pop-up will be used from old tool.

10.2 Electrical controls & IT

Electricals			
1	1	Field wiring, control & IT • Main Control Panel • Network Switches • Field Cabling • Barcode Scanner Controls • Conveyor Controls • IT • Control IT Software Package • NEOiT	Included

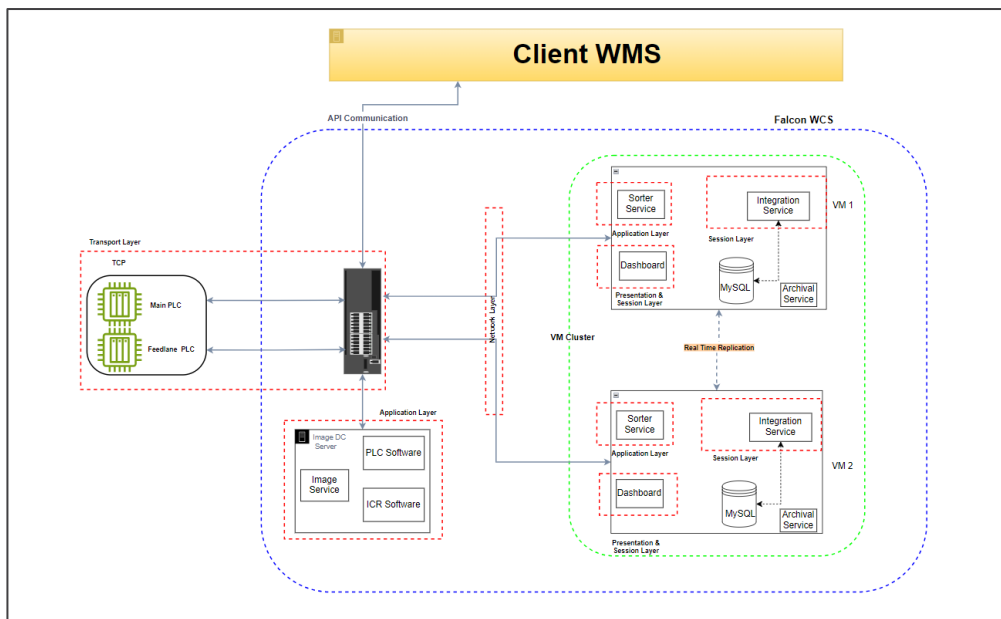
11. Falcon's WCS CONTROLIT

Falcon WCS (Warehouse Control System) is an in-house developed IT solution by Falcon Autotech, serving as the brain behind the company's sortation solutions. It manages the real-time movement of goods and data across the system, ensuring efficient operations in high-throughput warehouses. Falcon WCS integrates seamlessly with Warehouse Management Systems (WMS), Transport Management Systems (TMS), and other external applications via APIs to enhance operational efficiency.

11.1 System Architecture

High-Level Design (HLD) Overview

The Falcon WCS integrates with external systems like the Warehouse Management System (WMS) and Transport Management System (TMS). Communication occurs via **APIs/WSDL/ MQ Communication Protocol etc.** ensuring smooth data flow for order management, shipment tracking, and other critical operations.



Key Components:

- **Presentation and Session Layer:**
 - **MySQL Database:** Stores operational data, shipment details, and sortation instructions.
 - **Sorter Services:** Responsible for managing sorting logic and directing parcels to appropriate destinations.
 - **Dashboard:** Provides a user interface for real-time monitoring of warehouse operations and performance metrics.

- Integration Services: Handles communication with external systems (e.g., WMS, TMS) and ensures data consistency across platforms.

12.

- **Application Layer:**

- Image Services: Processes and manages images captured during the sortation process.
- ICR Software: Utilizes Image Character Recognition to read parcel labels and identify shipment information.
- PLC Software: Interfaces with Programmable Logic Controllers to manage the physical movement of parcels and control sortation equipment.

- **Transport Layer:**

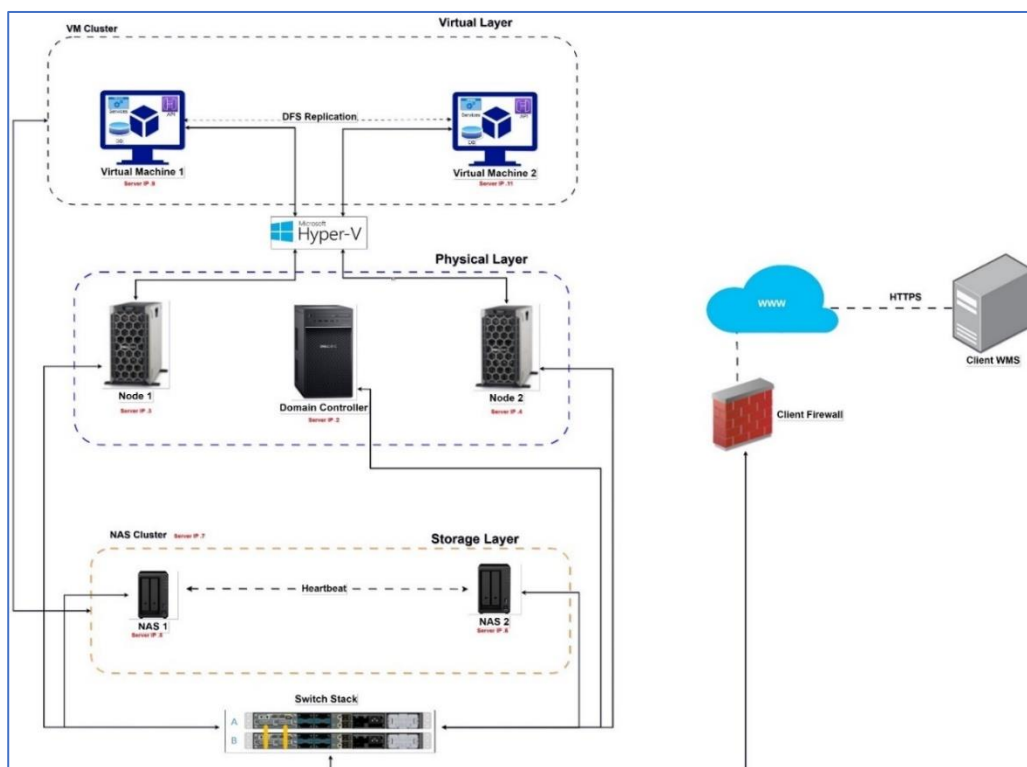
- Sorter PLCs: Receive commands from the session layer (Sorter Services) and execute sorting operations based on real-time data.

- **System Communication:**

All layers are connected via a Stacked Switch, which provides internet and intranet connectivity. Communication between the sortation system and external systems for results or shipment data occurs through this switch.

11.2 High Availability Architecture

The Falcon WCS architecture ensures uninterrupted operations using a **High Availability (HA) Server setup**. The system is designed to handle both planned and unplanned downtime, providing robust mechanisms for failover, replication, and data redundancy.



Key Components and Features of the High Availability Architecture:

1. Stacked Switch:

- Centralizes data exchange between **NAS, Nodes, Domain Controller (DC)**, and peripherals.
- Analyses packet headers to reduce unnecessary data transmission, enhancing LAN efficiency.

2. Domain Controller:

- **Heartbeat Monitoring:** Tracks the status of nodes and initiates VM failover when necessary.
- **Image Hosting:** Stores and manages images received from the ICR (Image Character Recognition).

3. NAS (Network Attached Storage):

- Centralized data storage providing access to connected devices and Virtual Machines.
- **Redundancy:** Two NAS boxes with mirrored drives ensure data protection and availability, offering a failsafe against hardware failure.

4. Node:

- **Hyper Terminals:** Nodes host and manage Virtual Machines (VMs) to run the warehouse control systems and related applications.
- **Clustering:** Nodes are clustered using **Microsoft Windows Cluster** to enable failover protection, ensuring continuous operation even in case of hardware failure.

5. Virtual Machine & InnoDB Cluster:

- **Primary VM:** Hosts Falcon WCS services, while a secondary backup on the node ensures failover through **Network Load Balancing (NLB)**.
- **InnoDB Cluster:** Ensures data replication using a **Master-Slave-Slave** setup for MySQL databases, maintaining consistency and availability.

6. NAS Cluster:

- **Unified File System:** NAS nodes share files across the cluster, ensuring no data loss during failover or disaster recovery.
- **Backup NAS:** Provides redundancy by replicating data between two NAS boxes, further safeguarding against failures.

Disaster Handling:

- **Recovery Time Objective (RTO) & Data Loss Objective (RPO):**
 - **VM Cluster Failure:** RTO = 1 hour; RPO = 1 hour.
 - **Node Failure:** No impact with a single failure; RTO = 4 hours if both nodes fail.
 - **NAS Failure:** Backup NAS available with no downtime, ensuring continued operation.

11.3 WCS User Interface

Overview

The Falcon WCS features a robust, user-friendly **Dashboard** that provides real-time visibility into warehouse and sortation operations. The Dashboard serves as the primary interface for monitoring key system metrics, tracking performance, and ensuring smooth operations.

Dashboard Overview

The WCS Dashboard offers real-time data visualization, helping warehouse operators and IT teams make data-driven decisions. Users can monitor system health, performance, and detect anomalies through an intuitive graphical interface.

Key Features of the Dashboard:

1. **System Health Monitoring:** Displays metrics such as CPU utilization, memory usage, disk performance, and system load across the infrastructure.
2. **Real-Time Sortation Monitoring:** Shows the real-time movement of parcels within the sortation system, including chute assignments and shipment statuses.
3. **Error Reporting:** Notifies users of system errors, network disruptions, and potential failures in real-time, allowing for quick resolution and minimal downtime.
4. **Performance Metrics:** Provides detailed reports on sortation throughput, parcel handling times, and system efficiency to ensure that warehouse targets are met.
5. **User Role Management:** The dashboard allows different levels of access based on user roles, ensuring that the right personnel can view or manage the system as needed.

In the context of this IT dashboard, the following user interactive screens are provided.

- **Dashboard (Home Screen):** Provides an overview of important metrics, data visualizations, and summary information related to the IT system or processes.



Figure 1 Dashboard

- **Live Bags:** Displays real-time information and status updates regarding bags or parcels currently in transit or being processed.

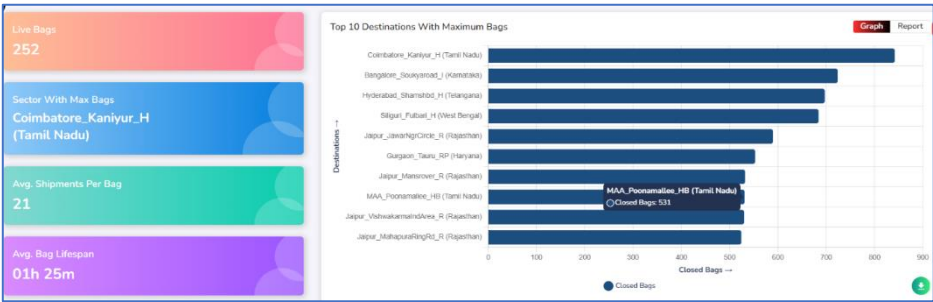


Figure 2 Live Bags

- **Bay Status:** Offers insights into the status and availability of different processing bays or areas within the system.

Bay Status				
Search Search Keywords Here...				
BAY ID	BAY STATUS	BAY TYPE	BAGGING MODE	SHIPMENTS SORTED IN BAY
Bay-1	ACTIVE	REGULAR	DIRECT	0
Bay-2	ACTIVE	REGULAR	DIRECT	0
Bay-3	ACTIVE	REGULAR	DIRECT	0
Bay-4	ACTIVE	REGULAR	DIRECT	0
Bay-5	ACTIVE	REGULAR	PTL	19
Bay-6	ACTIVE	REGULAR	DIRECT	0
Bay-7	ACTIVE	REGULAR	PTL	27
Bay-8	ACTIVE	REGULAR	PTL	7
Bay-9	ACTIVE	REGULAR	PTL	4
Bay-10	ACTIVE	UNHAPPY	DIRECT	0

Figure 3 Bay Status

- Search Processed Packages

Start Date *

3/2/2023

Start Time *

13:36

End Date *

3/2/2023

End Time *

13:51

AWB Number

Enter AWB Number Here ...

Package ID

Enter Package ID Here ...

Package Status *

All

Sorting Mode

All

Search

All Processed Packages

Search

Search Keywords Here ...

PACKAGE AWB	PACKAGE ID	BARCODES SCANNED	SCANNING MODE	INDUCTED LANE	BAY NUMBER	SORTING ERROR INFO	STATUS	PACKAGE LENGTH	PACKAGES
14908426909933	20842624	14908426909933	AUJ	F2	Bay-45	ACC	SUCCESS	32.4	28.1
14908427094112	20842626	--	AUJ	F2	Bay-24	ACC	SUCCESS	29.6	12.3
1490842613570663	20842626	1490842613570663	AUJ	F2	Bay-6	ACC	SUCCESS	29.3	12.7
1490842613011766	20842627	1490842613011766	AUJ	F2	Bay-305	ACC	SUCCESS	24.8	28.9
2649080652272	20842628	2649080652372_GFDN2JPF01017	AUJ	F2	Bay-28	ACC	SUCCESS	32.4	28.1
6963679551680	20842629	7581368537350FRRJGZGZGZ0839FA559R367553680	AUJ	F1	Bay-28	ACC	SUCCESS	33.5	44.3
14908421270814	20842630	--	AUJ	F2	Bay-60	DNF	FAULTED	24.5	20.6
906767647133	20842631	PHPP1574230289.593676847133_3C_547E299670DEJ.YK3D003089FA	AUJ	F1	Bay-28	ACC	SUCCESS	20.2	15.4
61281212009924	20842632	63648C120092.611230.1200924	AUJ	F1	Bay-16	ACC	SUCCESS	14.6	9.9
1490842498836	20842633	1490842498836	AUJ	F2	Bay-60	DNF	FAULTED	23.7	15.4

Items per page: 10

1 - 10 of 545

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>

- **Configuration Setting:** Enables users to configure and customize various settings and parameters within the IT system or dashboard.

NDC Management

Config Settings

RTVC Mode

Regex Pattern

Check NDC Code

NDC Code *

Enter NDC Code Here ...

Check NDC Code

Upload NDC



Select a file to upload
or drag and drop it here

Sample Td

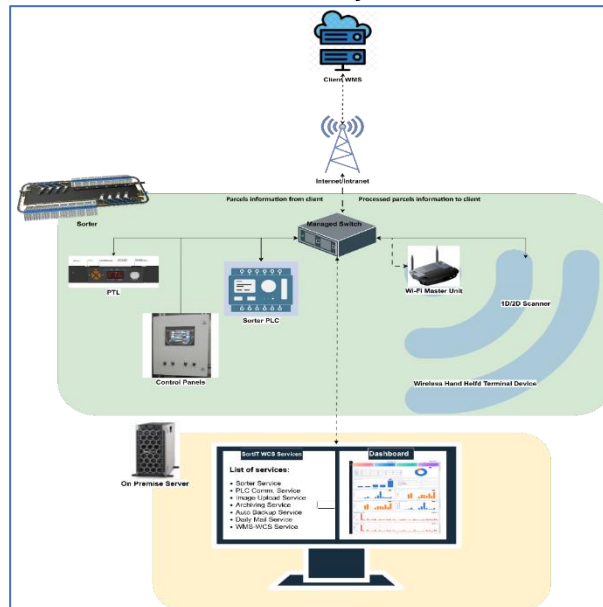
Upload

- **Report & Analysis:** Allows users to generate and access comprehensive reports, analytics, and insights based on the data collected by the IT dashboard.
- **Rejection Bay Mapping:** Provides functionality to map and manage rejection bays or areas where packages are deemed unsuitable for processing.
- **Alarms:** Displays alerts, notifications, or alarms related to system events, errors, or anomalies that require attention or investigation.
- **Calibration Settings:** Allows users to adjust and calibrate system settings, parameters, or sensors to ensure accurate and reliable performance.
- **Operator Management:** Offers features and tools to manage and monitor the operators or personnel responsible for operating the IT system.
- **User Management:** Provides functionality to manage user accounts, permissions, roles, and access levels within the IT dashboard.
- **User Guide:** The "User Guide" page offers comprehensive documentation and instructions on how to use the IT dashboard effectively. It serves as a reference guide for users.

11.4 Communication Architecture

Overview

Falcon WCS operates within a highly interconnected system, ensuring seamless communication between the WCS server, on-premises devices (such as Sorter PLCs, PTL devices, 1D scanners, and HHT devices), and client systems. This communication architecture facilitates real-time data exchange and operational control, optimizing sortation processes and warehouse efficiency.



On-Premises Communication

Sorter PLC Devices:

- **Protocol:** Falcon WCS communicates with Sorter PLCs using either the **Siemens S7** protocol or the **Omron communication** protocol.
- **Functionality:** The Sorter PLC devices receive sortation instructions from the WCS and execute the sorting process by directing parcels to the appropriate chute based on the system's real-time data.

PTL (Pick-to-Light) Devices:

- **Protocol:** PTL devices communicate with Falcon WCS using the **TCP/IP protocol**.
- **Functionality:** The system sends commands to the PTL devices for guiding manual picking operations by lighting up indicators at the appropriate bins or shelves, improving operational accuracy and speed.

1D Scanners:

- **Protocol:** These barcode scanners also use the **TCP/IP protocol** to communicate with the WCS.
- **Functionality:** The scanners capture barcode data from the parcels, and this information is sent to the WCS for processing, such as determining sorting destinations.

HHT (Handheld Terminal) Devices:

- **Protocol:** The wireless **HHT devices** communicate with Falcon WCS over **Wi-Fi**.
- **Functionality:**
 - The HHT devices send scan input data (e.g., barcodes) to the server over Wi-Fi.
 - The WCS processes this data and sends the required output instructions back to the HHT device and associated PTL devices.
 - The HHT device executes these instructions, facilitating real-time decision-making and execution for operators.

11.5 NEOiT

NEOiT is the brain of the NEO system. It is an in-house software stack to support all NEO operations, from order management to bot's route planning to bin allocations. Using robust AI algorithms and complex programming rules, NEOiT does all these operations like a breeze. Using API web services, it can quickly integrate with your WMSs, and with an intuitive user interface, it ensures faster and more efficient order preparations. NEOiT will ensure that the parcels storage inside the grid is based on the defined SLAs for shorter bot trips.



11.6 Client Communication

Data Transfer Methods:

- **API:** Falcon WCS can communicate processed data to client systems through **API** calls, allowing for seamless integration with external software.
- **MQ (Message Queuing):** Falcon WCS can also send data via **message queues**, ensuring reliable delivery of messages even during network downtime.
- **WSDL/XML:** For structured data exchanges, Falcon WCS supports **WSDL** and **XML** formats for client communication.
- **Other Protocols:** Additional methods for data transfer may include customized protocols depending on client requirements.

Purpose:

- The data sent to the client can include sortation results, system performance reports, and operational analytics, which can be used for further processing or reporting within external systems like **Warehouse Management Systems (WMS)** and **Transport Management Systems (TMS)**.

11.7 HAA Server Specifications

SN	Description	Qty
20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40		
1	Synology_storage_DS723+ 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,	2
	2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers.	
2	Dell Tower Model T440 -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4x32GB RDIMM Up to 8 3.5" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in	2
	Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	
3	Dell PowerEdge T40 Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T, Turbo,71W, TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM, DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3	1
	Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Net-gear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Converter Cadyce Brand	1

12. Key Components Make

Items	Make
Rollers	Falcon
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schneider/Equivalent
Bearings	NTN/SKF/Equivalent
HMIs	Siemens
MDR	Pulse/ Itoh Denki

13. Program Organisation

13.1 Program/Project Schedule

Go live will be end of march.

13.2 Program Management

Falcon Project management methodology constitutes of-

1. Creation and monitoring of the project plan.
2. Communication with Style Union Group.
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

13.3 Project Management

Upon awarding the contract, a dedicated Project Team is assigned to manage and execute the entire project from order receipt to customer acceptance testing and launch support.

The Project Manager/Project Engineer acts as the main contact with your team. Their responsibilities include:

- Coordinating the development of system and equipment design specifications.
- Managing project planning, scheduling, budgeting, reporting, and contract administration.
- Planning and allocating resources.
- Selecting and oversee equipment subcontractors.
- Preparing and reviewing the detailed Project Schedule with the Unilever Project Manager.
- Updating the Project Schedule as needed to identify critical activities and ensure proper implementation.

13.4 Systems Engineering

Experienced Systems Engineers handle all technical aspects of the project to ensure compliance with requirements. Their duties include:

- Leading the technical design effort.
- Defining system specifications.
- Coordinating engineering activities.
- Directing system test phases.
- Ensuring compliance with customer requirements and standards.
- Developing a Description of Operation for the project.

13.5 Site Management

The Site Manager oversees on-site activities of Falcon Autotech's personnel and subcontractors. Responsibilities include:

- Supervising equipment installation on-site.
- Coordinating subcontractor installation crews.
- Enforcing project specifications and drawings.
- Ensuring compliance with safety and plant standards.
- Communicating daily with customer representatives.
- Directing installation, checkout and testing.
- Reporting to the Project Manager/Project Engineer.

13.6 Software Engineering

Software Engineers are responsible for:

- Designing and developing material handling control system software applications.
- Installing and integrating computer systems and software with field equipment.
- Performing local module testing.

13.7 Electrical Controls Engineering

The Electrical Controls Engineer supports system startup and acceptance by:

- Determining control hardware requirements.
- Developing control software according to specifications.
- Creating project drawings for electrical controls.
- Testing PLC software in laboratory emulation before site delivery.

13.8 Mechanical Engineering

The Mechanical Engineer designs the mechanical components of the project. Responsibilities include:

- Developing and managing all mechanical drawings.
- Writing specifications for purchased equipment.

14. Commercial Terms

14.1 Price Sheet

Sr. No	Component	Qty per set	Set	UOM
1	NEO	1	1	Set
2	Conveyors	1	1	Set
3	Toolkit	1	1	Set
4	Safety Architecture Std	1	1	Set
5	Software Packages	1	1	Set
6	Installation & Commissioning	1	1	Set
7	Packaging & forwarding	1	1	Set
8	Project Management and Engineering Charges	1	1	Set
	Final discounted price (INR): ₹ 4,46,95,894			

* Note – Trays will be provided on a rental basis.

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges – Client's scope.
- Taxes and duties shall be extra as applicable.
- Including Packing charges
- Installation and Commissioning: Included

*Material Unloading & Laydown area is in customer scope.

14.2 Payment Terms

The proposal is based on the following payment schedule:

Payment Percentage	Stage
20%	Advance along with LOI/ PO
20%	After DAP Completion
40%	Before Dispatch
10%	Against Installation
10%	Against Handover

15. Warranty & SLA

Falcons offered System comes with a standard warranty of 1 year for complete system against manufacturing defect. Any damage caused to the system by an external source, abnormal use of equipment or any other factor not attributable to Falcon Autotech shall not be covered under warranty. Warranty period will start once the system is live and NOON GROUP starts using the system for its day-to-day operations.

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.

- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

15.1 Supply and Service Terms

The supply of material will be done by Falcon Autotech through its manufacturing unit in India. The installation and commissioning of the system will be done by Falcon along with our local partner.

The maintenance of the system during the warranty period will be done by Falcon. Any service visit required during the warranty period due to any manufacturing defect in the system will be done by Falcon. In case a spare part needs replacement while the system is in the warranty period, the same will be supplied and replaced by Falcon FOC. However, NOON GROUP is advised to keep critical spares parts recommended by Falcon in its own stock so that the same can be used without any delays. During the warranty as well as if the system is under CAMC post the expiry of the warranty period the same will be replenished by Falcon FOC.

Post warranty Falcon offers various service packages:

- Extended Warranty
- Consumable package
- Service Package
- On-site Support
- Comprehensive AMC

16. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Server PC
- Steelworks if not specified
- Consolidation Racks
- AWB Stations
- Printers, Bar code printers wherever required,
- Packing Tables and its associated accessories.
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Pull Cord Switches
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices.
- Traffic and route markings (outside FAPL storage grid)
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms if not mentioned in the BOM.
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.